Socio-Demographic Determinants of Health Insurance Costs in the United States

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Abstract. Health insurance pricing in the United States relies on a limited set of observable individual characteristics, yet how these variables actually shape cost differences is still not well understood. In this study, I apply a multivariate modeling approach to a dataset of US health insurance beneficiaries to identify and quantify the socio-demographic determinants of annual medical charges. I show that smoking status is the dominant cost driver, and that its interaction with body mass index creates a non-linear amplification effect that is entirely invisible in univariate analysis. Three predictive specifications are compared on a held-out test set: an OLS regression on log-transformed charges with HC3 robust standard errors, a Gamma generalized linear model with log link, and a Random Forest. The two linear specifications turn out to be empirically equivalent, while the Random Forest delivers a statistically significant gain in predictive accuracy that is confirmed by a paired bootstrap test. Building on these findings, I propose a five-tier risk scoring framework that translates statistical results into an operationally actionable pricing segmentation tool, and discuss its implications for actuarial practice and regulatory compliance.

1. Introduction

The United States health insurance market is characterized by significant heterogeneity in medical costs across insured individuals. This disparity is not random: it reflects differences in lifestyle, demographics, and health risk factors that the insurance pricing industry has long worked to measure. While the general association between smoking, obesity, and higher healthcare costs is well established in the medical literature, the relative contribution of each factor and how they interact remains poorly captured in predictive models built from standard administrative variables.

This study investigates a fundamental question that sits at the intersection of actuarial science and data science: to what extent can the annual medical charges of a US health insurance beneficiary be predicted from a small set of observable socio-demographic characteristics, and which of those characteristics matter most? Using a public dataset of 1,337 insured individuals after removing duplicates, I build three complementary predictive models, evaluated on a stratified hold-out test set, and benchmark them against a constant baseline. The first model is a multiple linear regression on log-transformed charges with HC3 robust standard errors, designed to provide interpretable coefficient estimates that can directly inform pricing decisions. The second is a Gamma generalized linear model with log link, the canonical actuarial specification for non-negative right-skewed cost data, included as a methodological reference point. The third is a Random Forest regressor, which relaxes the linearity assumption and captures complex interaction effects that the linear models can only approximate through manually specified interaction terms.

A central hypothesis motivating this work is that the effect of body mass index on medical charges is not independent but conditional on smoking status. A non-smoker with high BMI may generate only marginally higher costs than a non-smoker with normal BMI, while the same increase in BMI for a smoker may trigger a disproportionate amplification of costs. This interaction, if confirmed, has direct implications for how insurers should design their risk models, and it can only be revealed through multivariate analysis that explicitly accounts for the joint effect of these two variables. A second hypothesis, more methodological, is that the standard actuarial Gamma specification will not deliver a meaningful gain over OLS on log-transformed charges, and that any meaningful improvement in predictive accuracy must come from relaxing linearity altogether.

The remainder of this report is organized as follows. Section 2 presents the data and the main exploratory findings that motivate the modeling choices. Section 3 describes the three models, their results, and a formal comparison of their predictive performance through a paired bootstrap test on the test-set mean absolute errors. Section 4 discusses the implications of the findings, proposes operational recommendations, and acknowledges the limitations of the analysis. The Appendix provides the full OLS output, residual diagnostics, the Gamma GLM coefficient table, and additional model validation plots.

2. Data and Exploratory Analysis

2.1 Data

The analysis is based on the Medical Cost Personal Dataset, publicly available on Kaggle, which contains records for 1,338 insured individuals in the United States. For each individual, seven variables are available: age, sex, body mass index, number of children covered by the insurance contract, smoker status, geographic region, and the annual medical charges billed in US dollars. The target variable is charges, which represents the actual medical costs incurred rather than the insurance premium paid. The dataset contains no missing values, but a single exact duplicate (rows 195 and 581) is identified and dropped, leaving an effective sample of 1,337 observations. The four geographic regions are approximately balanced in terms of sample size. Table 1 provides a summary of the continuous variables.

Table 1. Descriptive statistics of continuous variables ($n = 1{,}337$)

Variable	Mean	Median	Std	Range
Age (years)	39.2	39.0	14.0	18–64
BMI	30.7	30.4	6.1	15.9–53.1
Children	1.1	1.0	1.2	0–5
Charges ($)	13,270	9,382	12,110	1,122–63,770

Before any modeling, the deduplicated dataset is split into a training set of 1,069 observations and a held-out test set of 268 observations, stratified by smoking status to preserve the 20.5% smoker prevalence in both partitions. All exploratory analysis, model fitting, and hyperparameter tuning are performed on the training set; the test set is touched only once at the model comparison step. Two features of the sample are worth highlighting before moving to the analysis. First, smokers represent only 20.5% of the training sample, which makes the dataset imbalanced with respect to the variable that will turn out to be the most influential. Second, the distribution of charges is strongly right-skewed, with a mean of $13,270 that is substantially higher than the median of $9,382. This asymmetry reflects the concentration of high costs among a minority of individuals and is a key reason why a log transformation is applied before fitting the OLS model.

2.2 Exploratory Findings

Figure 1 presents the main bivariate relationships between charges and the explanatory variables, computed on the training set only. Several patterns emerge from this visualization that directly motivate the modeling choices made in Section 3.

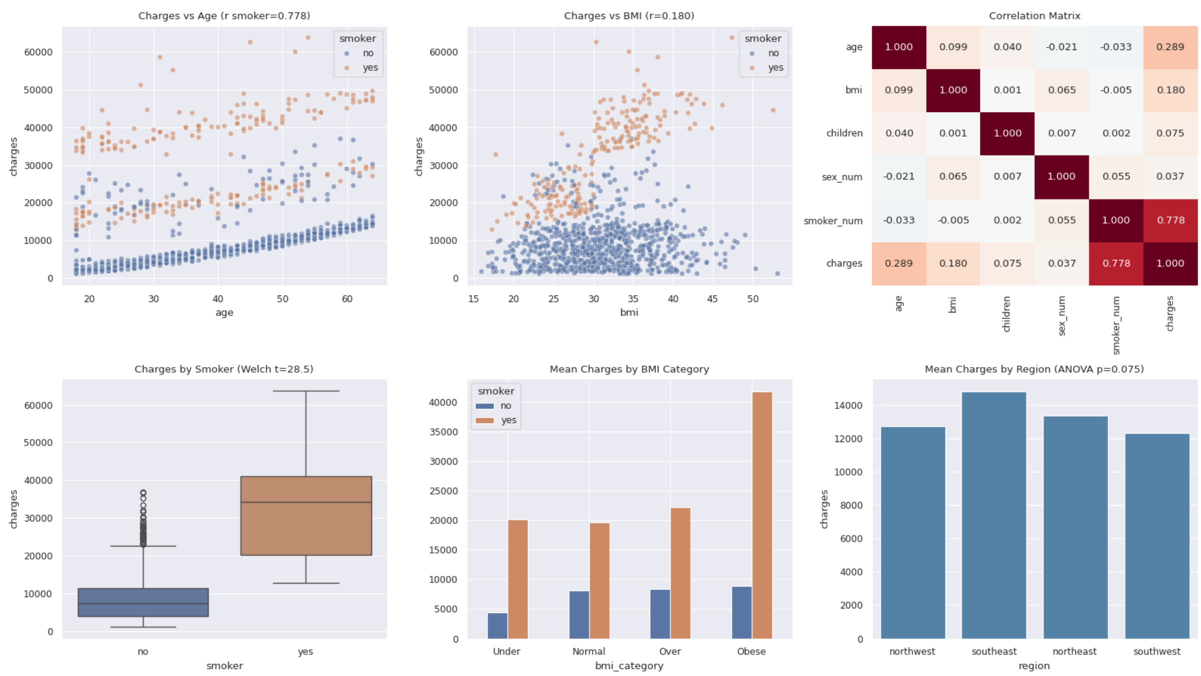

Figure 1. Bivariate relationships between charges and explanatory variables on the training set. Observations are color-coded by smoking status, with orange indicating smokers and blue indicating non-smokers. The scatter plots reveal that while age increases charges progressively for both groups, the BMI effect is concentrated almost entirely among smokers, pointing to a strong interaction between these two variables. The correlation matrix confirms that smoker status has by far the highest linear association with charges ($r = 0.78$), an order of magnitude above all other predictors.

The most striking pattern in the data is the separation between smokers and non-smokers. While age increases charges progressively for both groups, the overall level of costs for smokers is systematically higher across all age brackets. A Welch t-test formally confirms this difference, yielding a test statistic of 28.50 with a p-value effectively equal to zero, and smokers pay on average $31,938 compared to $8,568 for non-smokers, a ratio of 3.73 to 1. The geographic region, by contrast, does not reach conventional statistical significance on the training subsample, with an ANOVA p-value of 0.075 reflecting the loss of power from restricting the analysis to 1,069 observations.

Equally important is what the data reveals about BMI. In isolation, the Pearson correlation between BMI and charges is only 0.180, suggesting a weak association at best. But this aggregate figure masks a fundamentally different story: when the data is disaggregated by smoking status, it becomes clear that BMI almost exclusively amplifies costs for smokers. Non-smokers with high BMI do not exhibit systematically higher charges than their normal-weight counterparts, while obese smokers reach mean charges above $40,000 in the bivariate visualization. This observation is what motivates the inclusion of the "bmi_smoker" interaction term in the regression models, and it represents the central empirical finding of the exploratory phase of this work.

3. Modeling

3.1 Model 1: OLS on log(charges)

The first model is a multiple linear regression estimated by ordinary least squares on the log-transformed target variable, with HC3 heteroscedasticity-robust standard errors. The log transformation is justified by the strong right skew of the raw charges distribution, which would otherwise violate the normality assumption of the OLS residuals. After transformation, the skewness coefficient decreases from 1.52 to -0.11, indicating a distribution that is approximately symmetric. Under this specification, each regression coefficient $\hat{\beta}_j$ is interpreted as the approximate percentage change in charges associated with a one-unit increase in the corresponding variable, holding all others constant. Dollar-scale predictions are obtained by simple exponentiation; the Duan smearing correction was evaluated but found to over-correct on this heteroscedastic dataset, as discussed in Section 4.3.

The feature set includes ten variables: age, sex, BMI, number of children, smoker status, three regional indicator variables derived from the geographic region field, and two interaction terms. The first interaction, "bmi_smoker", is the product of BMI and the binary smoker indicator. The second, "age_smoker", captures the possibility that the age effect on charges differs between smokers and non-smokers. The model is estimated on the training set of 1,069 observations and evaluated on the test set of 268 observations. Stratified five-fold cross-validation on the training set returns a log-scale R^2 of 0.791 ± 0.105, the relatively high standard deviation reflecting the sensitivity of the OLS to the smoker composition of each fold.

The model achieves an R^2 of 0.806 on the training log target and 0.897 on the test log target. The slightly higher test R^2 reflects the specific stratified split used here rather than overfitting, which would push the test score below the training score. On the dollar scale, the model reaches an R^2 of 0.807 with a mean absolute error of $2,727 and a root mean squared error of $5,280. Table 2 summarizes the most informative coefficients, ranked by their standardized values to allow comparison of relative importance across variables measured on different scales.

Table 2. OLS key results on log(charges), ranked by standardized coefficient

Variable	$\hat{\beta}$	β^*	VIF	p-value
bmi_smoker	0.053	0.68	28.30	< 0.001
age	0.041	0.57	9.51	< 0.001
age_smoker	-0.034	-0.56	10.61	< 0.001
smoker_num	1.197	0.48	31.64	< 0.001
children	0.114	0.14	1.84	< 0.001
bmi	0.001	0.00	13.10	0.765

MAE = $2,727 · RMSE = $5,280 · High VIF on interaction terms is expected and acceptable for prediction.

The results in Table 2 confirm the central hypothesis of this study. The interaction term bmi_smoker is the strongest predictor in the model on the standardized scale, with $\beta^* = 0.68$ and a p-value effectively equal to zero. More strikingly, The raw BMI variable becomes completely non-significant when the interaction is included (p = 0.765), meaning that BMI has no detectable independent effect on charges once we account for the joint effect with smoking. This finding directly validates the hypothesis that BMI matters only in the context of smoking, and it would have been missed entirely by a model without the interaction term. The raw smoker indicator does carry the largest unstandardized coefficient ($\hat{\beta} = 1.197$), translating into a 231% increase in charges for smokers all else equal, but its standardized contribution falls behind "bmi_smoker" because the binary indicator varies on a narrower scale than the BMI-smoker product across the sample.

Age emerges as the second most important predictor on the standardized scale, contributing an estimated 4.1% increase in charges per additional year of age. The negative coefficient on "age_smoker" indicates that the age effect is somewhat attenuated for smokers, possibly because the baseline cost for smokers is already so high that the incremental contribution of aging is proportionally smaller. The number of children covered by the contract also has a significant positive effect, with each additional child associated with approximately 11.4% higher charges. The Variance Inflation Factors reported in Table 2 are elevated for the interaction terms and their components, an expected consequence of including both a variable and its interaction with another variable in the same model. This multicollinearity does not bias predictions and is acceptable here because the main goal of this analysis is prediction rather than causal inference on individual coefficients. The full OLS output with HC3 standard errors is provided in Appendix A, and the residual diagnostics are discussed in Appendix B.

3.2 Model 2: Gamma GLM with Log Link

The second model is a generalized linear model from the Gamma family with a log link function, fitted directly on the raw dollar-scale charges. This specification is the standard actuarial choice for cost data that is strictly positive and right-skewed, and its multiplicative structure mirrors how insurers actually price risk in practice. Unlike the OLS model on log-transformed charges, the Gamma GLM models the dollar-scale mean directly, so no retransformation correction is required.

The feature set is identical to that of Model 1, including the two interaction terms. On the test set, the Gamma GLM reaches an R^2 of 0.815, a mean absolute error of $3,100, and a root mean squared error of $5,169. The OLS and Gamma specifications turn out to be empirically equivalent on this dataset: OLS achieves a lower MAE ($2,727 versus $3,100) but a higher RMSE ($5,280 versus $5,169), and the explained variance is essentially indistinguishable between the two. This equivalence is informative in its own right and is discussed further in Section 4.3. The Gamma GLM coefficient table is reported in Appendix C.

3.3 Model 3: Random Forest

The third model is a Random Forest regressor, an ensemble of decision trees that can capture non-linear relationships and complex interactions without requiring them to be specified in advance. The Random Forest is estimated on the same training set used for the linear models, with hyperparameters selected through a grid search using five-fold cross-validation, stratified by smoking status. The search returns an optimal configuration of 200 trees, a maximum depth of 7 levels, and a minimum of 10 observations per terminal leaf. The same ten features as the linear models are used, including the manually specified interaction terms, to ensure a fair comparison even though a tree-based model could in principle discover the interactions on its own.

Figure 2 reports the permutation importance of each feature, computed on the test set with 20 repetitions to assess the contribution of each variable to predictive accuracy in a manner that is robust to the bias of standard impurity-based importance scores toward continuous variables with many distinct values.

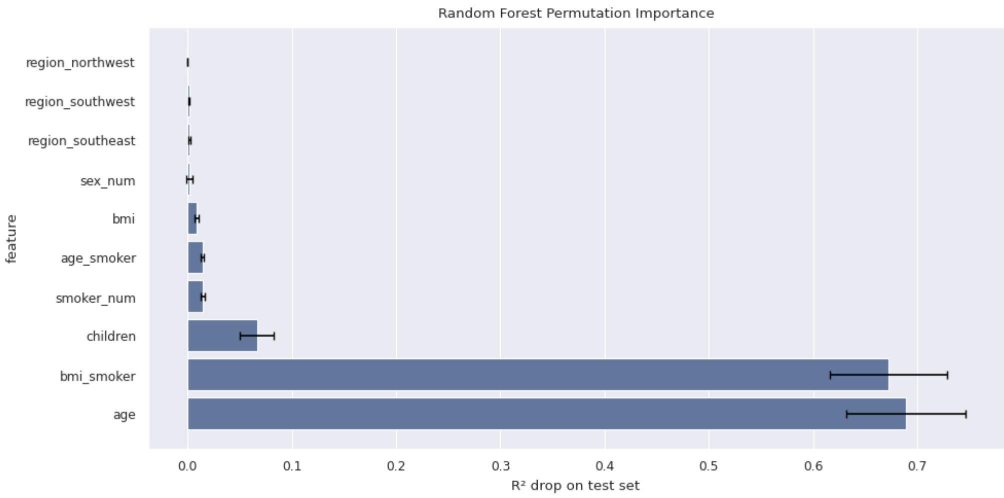

Figure 2. Random Forest permutation importance on the test set, measured as the drop in R^2 when a feature is randomly permuted. Error bars represent one standard deviation over 20 permutation repetitions. Age and the BMI-smoker interaction dominate the ranking almost equally, while the raw smoker indicator falls far behind because its information is already carried by the interaction term.

The Random Forest achieves an R^2 of 0.926 and a mean absolute error of $1,637 on the test set, compared to $2,727 for the OLS model, representing a 40% reduction in prediction error. The permutation importance rankings deserve a careful interpretation. "age" (importance 0.689) and "bmi_smoker" (0.673) dominate the ranking almost ex aequo, well ahead of "children (0.066) and "smoker_num" (0.015). The dramatic drop in the importance of the raw smoker indicator when measured by permutation is not a contradiction with the linear results but a consequence of the redundancy between "smoker_num" and "bmi_smoker" : when "smoker_num" is randomly permuted, the Random Forest can still reconstruct the smoker information through the interaction term, and the predictive performance is therefore barely affected. Permutation importance measures conditional contribution given the other features, not intrinsic importance, and this nuance is developed further in Section 4.3. Learning curves provided in Appendix D confirm that the model generalizes well, with a final gap of 0.04 between training and validation R^2.

3.4 Model Comparison

Table 3 brings together the performance metrics from all three predictive models and the constant baseline that predicts the training mean for every observation.

Table 3. Model performance summary on the held-out test set ($n = 268$)

Model	R^2	MAE ($)	RMSE ($)
Baseline (mean predictor)	−0.00	9,081	12,013
OLS log(charges)	0.807	2,727	5,280
Gamma GLM (log link)	0.815	3,100	5,169
Random Forest	0.926	1,637	3,269

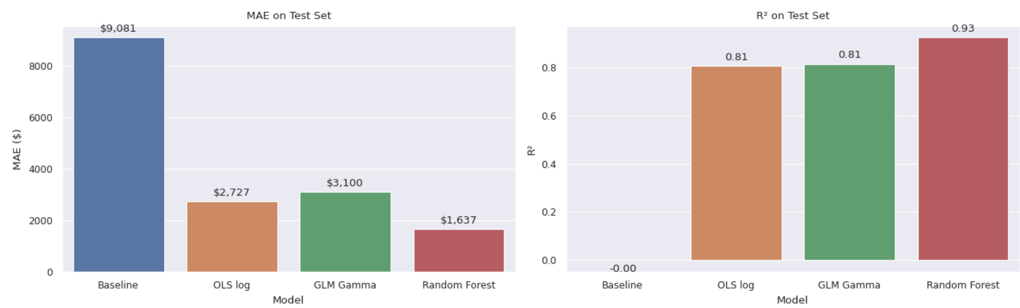

Figure 3. Visual comparison of MAE and R^2 between the four models. The Random Forest achieves a materially better fit on both metrics, reflecting its ability to capture the non-linearities and interactions that the linear models approximate but cannot fully represent. The two linear specifications are essentially indistinguishable.

To formally assess whether the gap between the OLS and the Random Forest is statistically meaningful rather than a product of sampling variability, I apply a paired bootstrap on the test-set absolute errors with 10,000 resamples. The procedure draws bootstrap samples of size $n = 268$ from the test set with replacement, computes the difference in mean absolute error between OLS and Random Forest on each resample, and constructs a 95% percentile confidence interval on the resulting distribution.

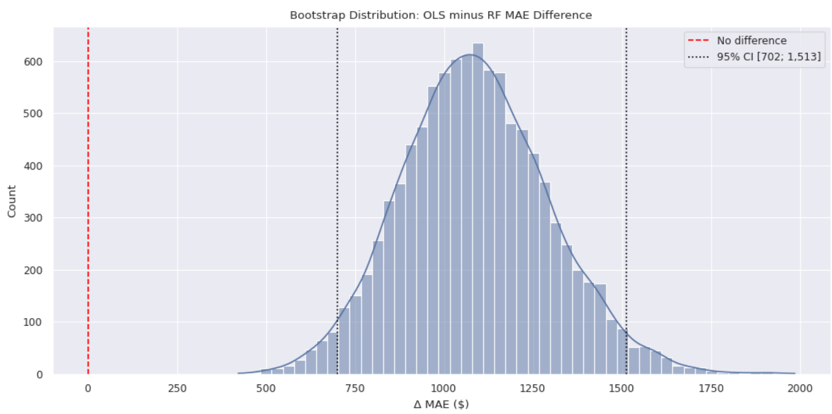

Figure 4. Bootstrap distribution of the OLS minus Random Forest MAE difference on the test set, with 10,000 paired resamples. The vertical dotted lines mark the 95% percentile confidence interval, both endpoints strictly positive, providing statistical evidence that the Random Forest delivers genuinely superior predictions.

The mean difference is \$1,089 in favor of the Random Forest, with a 95% confidence interval of [\$702; \$1,513]. The interval excludes zero, confirming that the Random Forest's advantage is statistically significant at the 5% level and operationally substantial in magnitude. The OLS and the Gamma GLM, by contrast, are not separated by any meaningful margin and should be considered functionally equivalent on this dataset.

4. Discussion

4.1 Summary of findings

This study set out to determine which socio-demographic factors drive health insurance cost disparities and whether these factors are sufficient to build accurate predictive models. The answer to both questions is affirmative, and the results are remarkably consistent across all three predictive models estimated in this analysis.

Smoking status is unambiguously the most important determinant of annual medical charges. The Welch t-test, the OLS coefficients, the Gamma GLM coefficients, and the Random Forest permutation importance all lead to the same conclusion: smokers generate healthcare costs that are systematically and much higher than those of non-smokers, regardless of age or BMI. But the more nuanced and methodologically interesting finding is that BMI does not operate as an independent risk factor. Its effect on charges is almost entirely mediated by its interaction with smoking status, an effect that is invisible in simple bivariate analysis ($r = 0.18$) and only detectable in a properly specified multivariate model where `bmi_smoker` emerges as the dominant standardized predictor while raw BMI is non-significant. This result has direct implications for how insurers should think about BMI as a pricing variable: it is not a risk factor in isolation but a risk amplifier conditional on smoking.

4.2 Risk segmentation and operational implications

Figure 5 illustrates a five-tier risk segmentation framework that translates the statistical findings of this study into an actionable tool for insurance pricing. Rather than using ad-hoc weights for each risk factor, the segmentation comes directly from the OLS model: the quintile thresholds of the OLS predictions on the training set ($4,050, $7,159, $11,114, $17,946) are frozen and applied to the test set predictions to assign each test individual to one of five tiers. This approach is monotone by construction, derived from a fitted model rather than expert opinion, and validated on data the model has not seen.

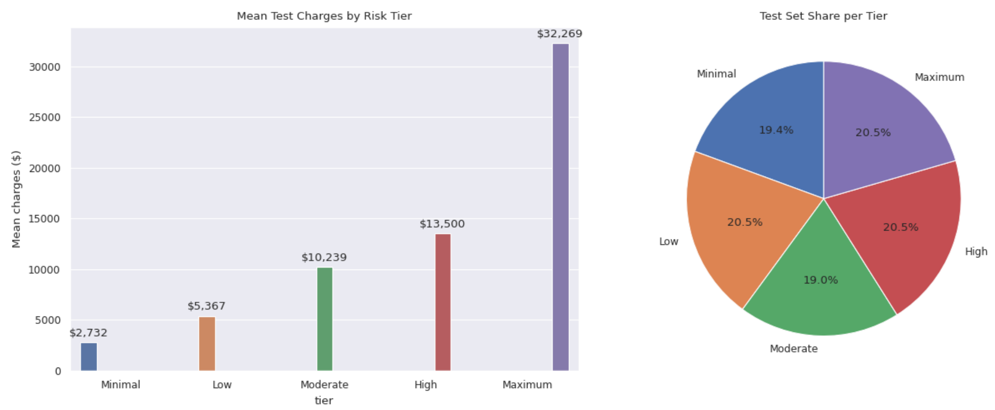

Figure 5. Risk profile segmentation based on the OLS prediction quintiles. The bar chart on the left shows the mean charges associated with each tier on the test set, ranging from $2,732 for the minimal risk group to $32,269 for the maximum risk group. The pie chart on the right shows that each tier captures approximately 20% of the test population by quintile construction, with the maximum tier concentrating the costliest 20% by predicted charges.

The 11.8-fold ratio between the highest and lowest cost tiers provides a statistically grounded justification for differentiated pricing. From an operational standpoint, these findings support several concrete recommendations. A five-tier pricing structure with spreads proportional to the observed cost differentials would better reflect the underlying risk distribution than a flat or two-tier system. A systematic surcharge for smokers calibrated to the observed $23,370 average cost differential would align premiums more closely with actual risk. The BMI-smoker interaction should be incorporated as a primary actuarial risk signal rather than treating BMI as a standalone factor. Finally, while the Random Forest achieves higher predictive accuracy, the OLS model is better suited for regulatory compliance under the interpretability requirements of US insurance regulation, and post-hoc explainability methods such as SHAP values could be integrated to bring the Random Forest into compliance for production deployment.

4.3 Methodological observations and limitations

Three methodological observations from this analysis deserve specific discussion as they shape the conclusions and point toward sound practice in similar studies. The first concerns the Duan smearing factor, traditionally applied to convert log-scale predictions back to the dollar scale under the assumption of homoscedastic residuals. On this dataset, the global Duan factor of 1.13 over-corrects substantially: applied to the OLS predictions, it degrades the test MAE from $2,727 to $3,687, because the heavy-tailed smoker residuals inflate the global correction factor and lead to systematic over-prediction for the 80% of non-smokers. A naive exponentiation of the log predictions turns out to be empirically near-unbiased on this dataset (mean predicted $13,029 against mean actual $12,973) and is therefore retained as the dollar-scale prediction strategy. This finding illustrates that theoretically motivated corrections should be validated against the empirical bias they are meant to address before being applied in production.

The second observation concerns the apparent contradiction between the OLS standardized coefficients and the Random Forest permutation importance. In the OLS, "bmi_smoker" ($\beta^* = 0.68$) leads "age" ($\beta^* = 0.57$) and "smoker_num" ($\beta^* = 0.48$); in the Random Forest, "age" (0.689) and "bmi_smoker" (0.673) are almost ex aequo while "smoker_num" drops to 0.015. The two metrics measure different quantities: the standardized linear coefficient quantifies the marginal effect of a variable at fixed scale, while permutation importance quantifies the conditional contribution given the other features in the model. When "smoker_num" is permuted in the Random Forest, the model can still recover the smoker signal through "bmi_smoker", which is why permutation collapses the apparent importance of "smoker_num" despite its central role in the data. Both metrics tell true but different stories about the same underlying structure.

The third observation is the empirical equivalence of the OLS log model and the Gamma GLM with log link. Actuarial theory generally favors the Gamma specification for cost data that is strictly positive and right-skewed, and its multiplicative structure mirrors insurance pricing practice. On this dataset, however, the two specifications produce essentially indistinguishable predictive performance, with OLS slightly better on MAE and Gamma slightly better on RMSE. The choice between them on this data should therefore be driven by interpretability and downstream consistency rather than by predictive gain. Only the Random Forest produces a statistically and operationally meaningful improvement, as the paired bootstrap test confirms.

Beyond these methodological points, the analysis is subject to standard limitations. The dataset covers only insured individuals, which introduces a selection bias toward people who already have access to the insurance market. The data is also cross-sectional, providing a snapshot rather than a longitudinal view of how an individual's costs evolve over time. Future work could address these limitations by incorporating longitudinal data, by extending the model to include behavioral and clinical variables not available in this dataset, and by integrating SHAP-based explainability into the Random Forest to meet the interpretability standards required by regulators.

4.4 Conclusion

This study demonstrates that a small set of socio-demographic variables is sufficient to explain about 90% of the variance in US health insurance charges on a held-out test set, provided that the interaction between smoking status and body mass index is properly accounted for. The convergence of results across three methodologically distinct models gives confidence that these findings reflect genuine structure in the data rather than artifacts of a particular modeling choice. The Random Forest improves the mean absolute error by $1,089 over OLS, a gap that is statistically significant at the 5% level via paired bootstrap (95% CI [$\$702;\ \$1{,}513$]) and operationally meaningful in a pricing context. The proposed five-tier risk segmentation framework, derived from the OLS prediction quintiles and validated on the held-out test set, provides a direct bridge between the statistical analysis and operational practice, delivering an 11.8-fold cost ratio between the top and bottom tiers and offering insurers a transparent and defensible tool for risk-based pricing.

A. OLS Full Summary

Table 4. Full OLS regression results on log(charges) with HC3 robust standard errors, training set $n = 1{,}069$

Variable	$\hat{\beta}$	Std Err	z	p-value
const	7.197	0.089	81.22	< 0.001
age	0.041	0.001	33.26	< 0.001
sex_num	-0.104	0.025	-4.10	< 0.001
bmi	0.001	0.002	0.30	0.765
children	0.114	0.010	11.48	< 0.001
smoker_num	1.197	0.127	9.42	< 0.001
region_northwest	-0.065	0.037	-1.72	0.085
region_southeast	-0.154	0.038	-4.04	< 0.001
region_southwest	-0.165	0.036	-4.57	< 0.001
bmi_smoker	0.053	0.004	14.14	< 0.001
age_smoker	-0.034	0.002	-20.12	< 0.001
$R^2_{\text{train}} = 0.806 \cdot R^2_{\text{test, log}} = 0.897 \cdot \text{CV } R^2 = 0.791 \pm 0.105$				

B. Residual Diagnostics

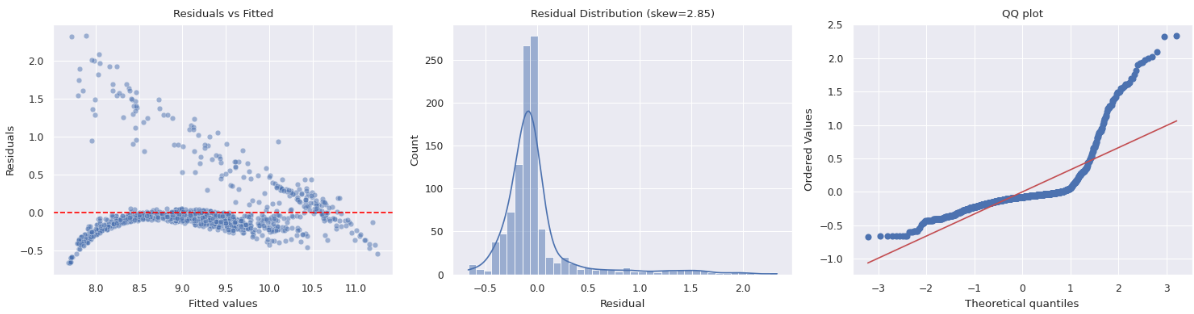

Figure 6. OLS residual diagnostics. The fan-shaped pattern in the residuals versus fitted values plot reveals heteroscedasticity, which is formally confirmed by the Breusch-Pagan test (statistic = 67.50, $p = 1.3 \times 10^{-10}$). The residual distribution and QQ-plot show significant departures from normality, with a skewness of 2.85 and a kurtosis of 9.48. Both violations are a direct consequence of the bimodal cost structure created by the smoker and non-smoker populations, and they motivate the use of HC3 robust standard errors throughout the OLS analysis.

C. Gamma GLM Results

Table 5. Gamma GLM with log link, full coefficient table, training set $n = 1{,}069$

Variable	$\hat{\beta}$	Std Err	z	p-value
const	7.668	0.147	52.02	< 0.001
age	0.033	0.002	18.56	< 0.001
sex_num	-0.087	0.044	-1.98	0.048
bmi	0.001	0.004	0.23	0.819
children	0.097	0.019	5.20	< 0.001
smoker_num	0.767	0.305	2.51	0.012
region_northwest	-0.065	0.063	-1.03	0.305
region_southeast	-0.154	0.063	-2.43	0.015
region_southwest	-0.192	0.063	-3.03	0.002
bmi_smoker	0.053	0.009	5.96	< 0.001
age_smoker	-0.026	0.004	-6.47	< 0.001
R^2_{test} = 0.815 · MAE = \$3,100 · RMSE = \$5,169				

D. Learning Curves

Figure 7. Random Forest learning curves showing train and validation R^2 as a function of training set size. The two curves converge steadily as the training size increases, with a final gap of 0.04 at full sample size (train $R^2 = 0.85$, validation $R^2 = 0.81$), confirming that the model generalizes well and is not memorizing the training data.